Quantification: 535_z13c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	28
interior lacuna count	24
frame-edge lacuna count	4

Interior lacunae only: mean, median and sample SD (n = 24)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2208.67	937.5	2477.05	px^2
major_axis_length_px	lacuna shape	75.7618	65.0635	44.8906	px
minor_axis_length_px	lacuna shape	34.4873	26.9966	19.5676	px
aspect_ratio	lacuna shape	2.2328	2.2373	0.6707	unitless
eccentricity	lacuna shape	0.8529	0.8944	0.1073	unitless
solidity	lacuna shape	0.7855	0.7833	0.0942	unitless
roots_count	headline	7.25	6	4.8477	count
ring_length_r30_px	headline	228.958	214.5	142.166	px
ring_length_r60_px	reported	453.917	410.5	255.587	px
owned_length_px	ownership-dependent	487.754	368.75	379.49	px
edge_count	ownership-dependent	23.8333	18	19.4928	count
mean_edge_length_px	ownership-dependent	21.9507	20.7427	8.6034	px
perimeter_px	normalised	236.303	171.146	174.159	px
ring_area_r30_px2	normalised	7843.88	5949	4157.46	px^2
ring_area_r60_px2	normalised	16321.8	15709.5	7436.18	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9981	1	0.0073	unitless
ring_density_r30	normalised	0.0285085	0.0288628	0.00916867	px^-1
ring_density_r60	normalised	0.027333	0.0284838	0.0078369	px^-1
roots_per_100px_perimeter	normalised	3.2848	3.0662	1.6042	per 100 px
ring_attached_length_r30_px	network v2	196.708	177.5	136.125	px
ring_attached_length_r60_px	network v2	335.333	322	245.461	px
ring_length_w_r30_px	network v2	258.096	239.095	161.914	px
ring_length_w_r60_px	network v2	512.1	461.369	289.783	px
sholl_crossings_r10	network v2	5.7083	5	4.0805	count
sholl_crossings_r20	network v2	5.7083	5.5	3.2098	count
sholl_crossings_r30	network v2	6.75	6	3.9037	count
loop_count	loops, report only	0.9583	1	0.8065	count
loop_edge_share	loops, report only	0.1232	0.1231	0.1143	share
ring_width_mean_r30_px	width	7.3066	7.5871	1.2719	px

Per field
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measure	value	units
analysed_area_px2	986148	px^2
total_skeleton_length_px	20596	px
canalicular_length_density_per_px	0.0208853	px^-1
junction_count	377	count
junction_density_per_px2	0.0003823	px^-2
lacuna_count	28	count

lacunae_per_px2	2.839e-05	px^-2
skeleton_component_count	344	count
mean_component_length_px	59.8721	px
median_component_length_px	20	px
field_length_density_w_per_px	0.0234981	px^-1
field_density_without_flagged_per_px	0.0204081	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	815	px
unattached_component_count	49	count
canalicular_length_density_attached_per_px	0.0208853	px^-1
field_length_density_with_unattached_per_px	0.0217117	px^-1
loop_count	27	count
loop_edge_share	0.0793	share
edge_count	1097	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.396	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	19257	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	520	50.9524	21.26	2.3966	0.9088	0.6061	0.134	15.2558	959.056	True
2	509	42.4483	19.3806	2.1902	0.8897	0.7313	1.0605	62.053	174.511	False
3	3449	94.2445	51.4301	1.8325	0.838	0.8008	-1.3442	87.5868	746.098	False
4	424	32.3309	18.5247	1.7453	0.8196	0.7837	-0.9381	141.42	858.868	False
5	434	32.4903	18.5735	1.7493	0.8205	0.7993	1.0377	151.09	693.419	False
6	418	45.2622	13.5025	3.3521	0.9545	0.8132	-0.8448	294.464	171.199	False
7	429	36.6586	16.6305	2.2043	0.8912	0.8684	0.0347	305.65	1016.41	True
8	1518	84.7212	28.0385	3.0216	0.9436	0.7609	0.4889	372.306	481.817	False
9	424	39.0325	14.1936	2.75	0.9315	0.9002	-0.6027	429.66	215.08	False
10	916	42.5998	32.8395	1.2972	0.637	0.7829	0.4554	434.044	377.415	False
11	6729	150.978	65.1171	2.3186	0.9022	0.7721	-0.3046	496.001	291.95	False
12	3320	105.162	44.1209	2.3835	0.9077	0.7721	-1.4464	464.635	577.918	False
13	3645	102.737	49.6538	2.0691	0.8754	0.8697	0.3597	580.157	513.438	False
14	2801	98.9222	38.3287	2.5809	0.9219	0.8173	0.524	580.027	151.776	False
15	9054	174.453	85.8111	2.033	0.8707	0.6586	0.1605	649.475	679.118	False
16	1209	70.3083	25.9548	2.7089	0.9294	0.8103	-1	737.055	638.021	False
17	3256	105.273	45.222	2.3279	0.903	0.7485	-1.2191	649.232	325.526	False
18	524	31.117	22.6042	1.3766	0.6872	0.8748	-1.3629	760.422	888.611	False
19	597	63.166	16.0382	3.9385	0.9672	0.6958	-1.4137	764.35	853.412	False
20	543	31.4075	23.6295	1.3292	0.6588	0.9282	0.087	780.65	298.357	False
21	455	27.4749	22.5328	1.2193	0.5722	0.9362	1.1384	784.457	318.629	False
22	7705	171.941	75.2669	2.2844	0.8991	0.7155	-1.3369	837.132	645.403	False
23	2929	114.614	39.6482	2.8908	0.9383	0.6816	1.0561	854.515	315.604	False
24	959	66.9611	35.0006	1.9131	0.8525	0.5464	-0.2618	924.19	350.901	False

25	580	52.7819	21.5044	2.4545	0.9132	0.716	1.2958	848.429	862.598	False
26	610	37.8566	20.7792	1.8218	0.8359	0.937	-0.7458	884.313	871.107	False
27	2259	90.1772	47.5611	1.896	0.8496	0.6829	0.6436	985.683	487.498	True
28	6212	244.279	36.0383	6.7783	0.9891	0.7991	-1.5538	1007.31	703.743	True

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	7	169	259	182.439	10	18.2439	5.8346
2	6	186	464	511.245	24	21.3019	7.0114
3	9	380	763	1034.62	50	20.6924	7.2345
4	2	249	605	229.853	17	13.5208	6.7267
5	4	140	265	330.492	15	22.0328	5.721
6	5	232	581	802.784	33	24.3268	7.8466
7	3	124	249	165.167	8	20.6458	7.3128
8	7	181	367	123.953	11	11.2685	5.2487
9	6	155	389	597.186	32	18.6621	9.0159
10	11	225	404	769.826	33	23.328	7.6
11	16	453	812	867.252	49	17.699	8.3931
12	7	261	359	225.024	12	18.752	6.6633
13	3	204	417	202.64	7	28.9485	8.5039
14	12	263	531	499.031	24	20.793	7.9127
15	13	518	1025	1538.8	68	22.6294	7.8684
16	5	72	91	211.409	4	52.8523	5.8056
17	13	390	559	403.948	18	22.4416	7.669
18	1	95	292	194.196	8	24.2745	7.5743
19	4	72	225	175.569	11	15.9608	6.8691
20	2	86	206	123.669	7	17.667	8.7709
21	3	75	138	391.836	20	19.5918	8.2757
22	17	514	1013	1167.06	74	15.7711	7.6634
23	14	339	523	677.286	29	23.3547	6.2411
24	9	277	496	345.664	18	19.2036	7.4029
25	4	73	172	269.51	7	38.5014	9.4276
26	1	55	197	13.2426	1	13.2426	3.9128
27	6	270	448	255.208	17	15.0122	8.2074
28	8	288	673	548.86	31	17.7052	8.3581

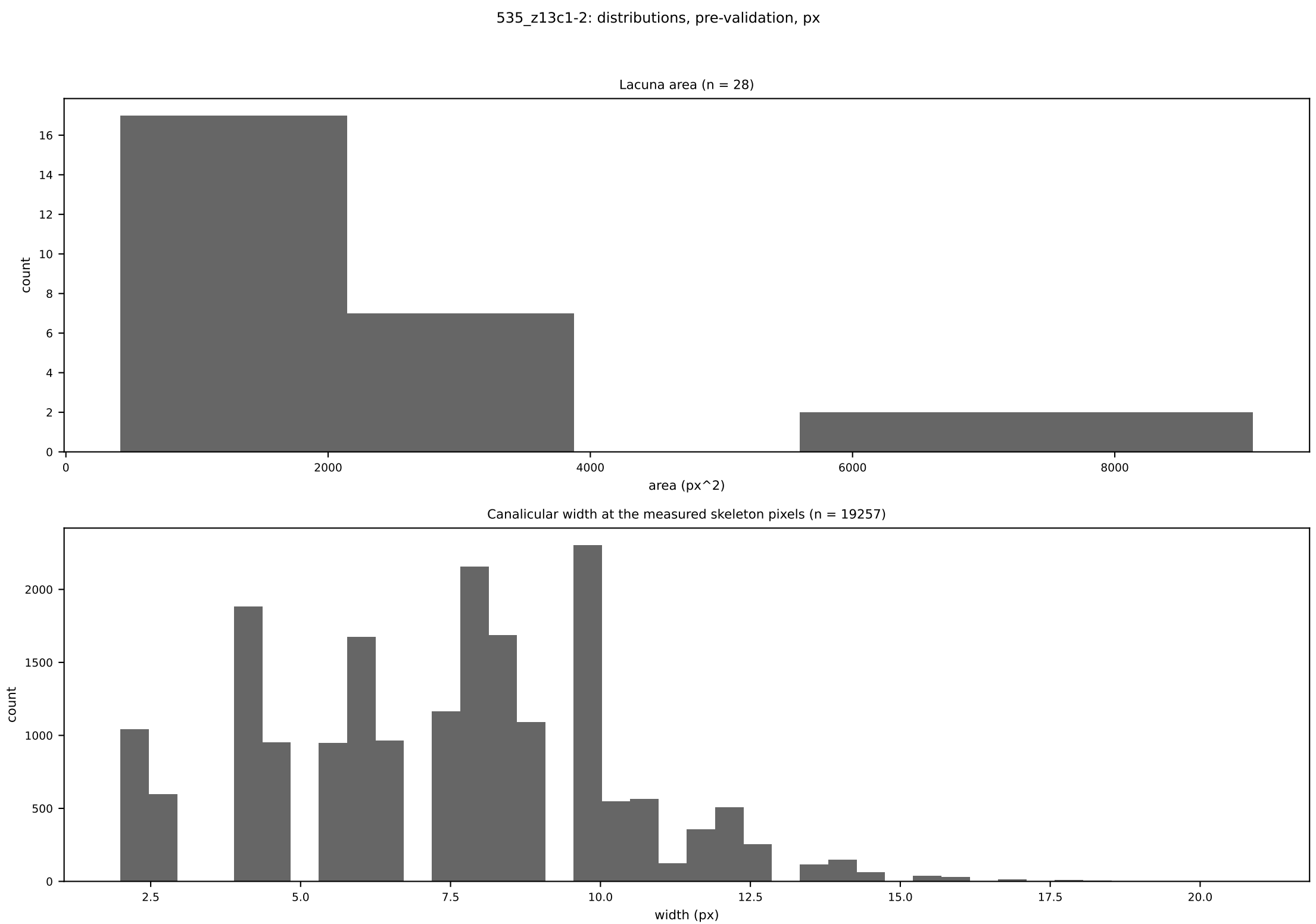
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	153.297	4649	11757	0.6846	0.6173	0.0363519	0.0220294	4.5663
2	125.882	6125	17204	1	0.9656	0.0303674	0.0269705	4.7664
3	311.848	10440	20760	1	0.9885	0.0363985	0.0367534	2.886
4	101.569	5469	14333	1	1	0.0455294	0.0422103	1.9691
5	104.225	4890	11875	1	1	0.0286299	0.0223158	3.8378
6	104.569	5737	17086	1	1	0.0404393	0.0340044	4.7816
7	96.9056	3199	9195	0.5807	0.5526	0.0387621	0.0270799	3.0958
8	215.622	8573	18773	1	1	0.0211128	0.0195494	3.2464
9	94.5685	5399	12868	1	1	0.028709	0.03023	6.3446
10	145.782	5773	12342	1	1	0.0389745	0.0327337	7.5455
11	433.103	13611	21359	1	1	0.0332819	0.0380168	3.6943
12	326.563	11088	20860	1	1	0.023539	0.01721	2.1435
13	270.936	10229	21506	1	1	0.0199433	0.0193899	1.1073
14	264.007	10205	23680	1	1	0.0257717	0.022424	4.5453
15	731.429	16860	32093	1	1	0.0307236	0.0319384	1.7773
16	194.894	5275	8641	1	1	0.0136493	0.0105312	2.5655
17	328.877	11820	23768	1	1	0.0329949	0.023519	3.9528

18	99.6335	3693	9318	1	1	0.0257243	0.0313372	1.0037
19	147.397	4813	9956	1	1	0.0149595	0.0225994	2.7138
20	96.0833	3059	6126	1	1	0.0281138	0.0336272	2.0815
21	85.734	2578	5393	1	1	0.0290923	0.0255887	3.4992
22	655.553	17714	33378	1	1	0.0290166	0.0303493	2.5932
23	380.475	9227	17435	1	1	0.03674	0.0299971	3.6796
24	212.208	6468	13647	1	1	0.0428262	0.036345	4.2411
25	139.498	4945	8638	1	1	0.0147624	0.019912	2.8674
26	100.811	4262	10684	1	1	0.0129047	0.0184388	0.992
27	275.764	8026	16014	0.7864	0.7193	0.0336407	0.0279755	2.1758
28	597.973	11032	21934	0.5965	0.5686	0.0261059	0.030683	1.3379

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	145	160	197.238	299.806	3	4	2
2	157	275	213.51	530.659	3	4	7
3	323	587	426.848	870.323	10	9	12
4	245	349	279.066	672.487	8	5	7
5	117	210	150.426	294.551	3	6	5
6	169	418	252.095	645.416	5	5	8
7	111	174	143.782	286.978	3	5	4
8	99	121	202.894	411.789	4	6	7
9	128	369	181.681	451.961	3	5	6
10	215	362	253.066	461.789	6	6	6
11	419	763	519.517	925.423	13	9	13
12	189	210	288.309	393.492	6	12	6
13	186	318	226.095	460.948	7	8	5
14	249	430	303.078	599.286	7	8	8
15	499	933	585.902	1161.42	14	12	14
16	68	87	83.2548	106.397	1	2	1
17	316	361	447.475	633.428	8	8	11
18	83	194	111.054	333.492	3	3	4
19	72	136	77.8701	245.368	2	1	1
20	63	118	94.1838	238.338	2	1	5
21	47	72	80.2843	148.083	1	3	2
22	487	866	576.718	1141.13	15	9	13
23	286	422	385.049	591.316	7	4	6
24	217	326	317.421	556.931	5	7	11
25	73	107	80.6985	193.125	4	2	2
26	14	14	57.799	223.238	0	2	2
27	254	354	304.522	501.789	7	9	7
28	243	422	315.865	751.027	7	10	8


535_z13c1-2: distributions, pre-validation, px
Lacuna area (n = 28)
count
area (px^2)
0
2
4
6
8
10
12
14
16
0
2000
4000
6000
8000
Canalicular width at the measured skeleton pixels (n = 19257)
count
width (px)
0
500
1000
1500
2000
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0